

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
SIMATS ENGINEERING
COMPUTER SCIENCE AND ENGINEERING
CSA17 - ARTIFICIAL INTELLIGENCE

Experiment No. 9: Python Program to Implement the Travelling Salesman Problem

Aim

To write a Python program to solve the Travelling Salesman Problem (TSP), which involves finding the shortest possible route that visits every city exactly once and returns to the starting city, using a brute-force permutation-based search.

Algorithm

1. Start.
2. Represent the cities and the distances between them using a distance matrix (or a dictionary of dictionaries), where `graph[city1][city2]` gives the distance between city1 and city2.
3. Read the starting city from which the salesman begins and ends the tour.
4. Generate all possible permutations of the remaining cities (excluding the start city) using `itertools.permutations`.
5. For each permutation, construct a complete tour by prefixing and appending the start city to the permutation, forming a closed loop.
6. Compute the total travel cost of this tour by summing the distances between consecutive cities in the route, using the distance matrix.
7. Compare the computed cost with the minimum cost found so far; if the current tour's cost is lower, update the minimum cost and store this tour as the best path.
8. Repeat steps 5-7 for every permutation of the remaining cities.
9. After all permutations have been evaluated, the stored best path represents the optimal (shortest) route.
10. Print the optimal route and its minimum total travel cost.
11. Stop.

Pseudocode

```
BEGIN
  INPUT: graph (distance matrix), start_city
  cities = all cities except start_city
  min_cost = infinity
  best_path = NULL

  FOR each permutation P of cities DO
    path = [start_city] + P + [start_city]
    cost = 0

    FOR i = 0 TO length(path) - 2 DO
      cost = cost + graph[path[i]][path[i+1]]
    END FOR

    IF cost < min_cost THEN
      min_cost = cost
    END IF
  END FOR
```

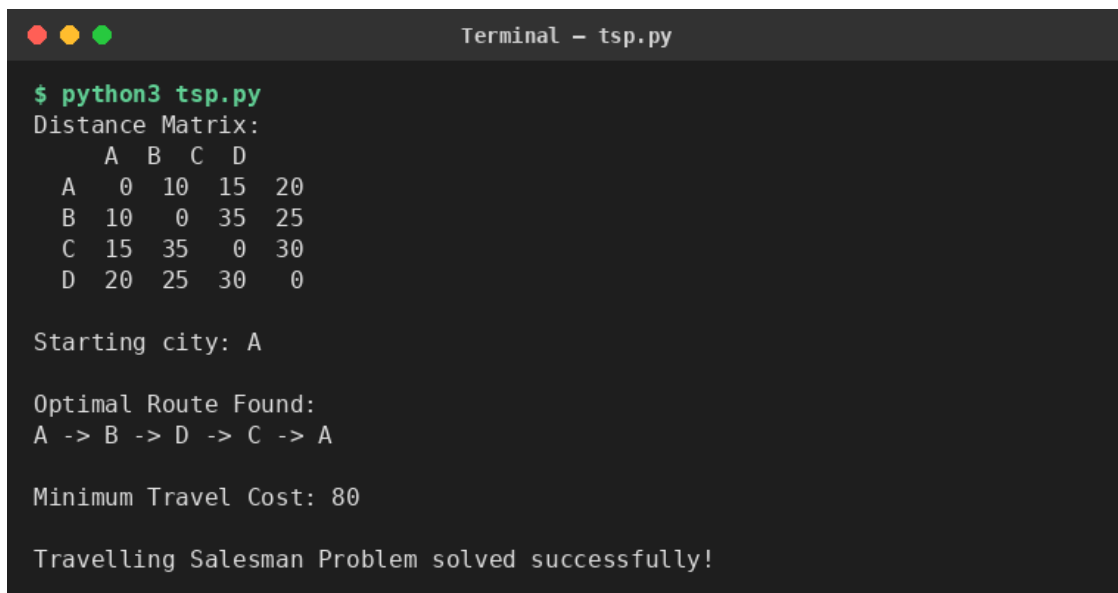
```
        best_path = path
    END IF
END FOR

PRINT best_path, min_cost
END
```

Result

The Python program for the Travelling Salesman Problem was executed successfully using the brute-force permutation approach. For four cities (A, B, C, D) with the given distance matrix, the program evaluated all possible tours starting and ending at city A, and identified the optimal route A -> B -> D -> C -> A with a minimum travel cost of 80. Thus, the Travelling Salesman Problem was solved successfully using Python.

Output Screenshot:



```
Terminal - tsp.py

$ python3 tsp.py
Distance Matrix:
  A  B  C  D
A  0  10 15 20
B  10  0 35 25
C  15 35  0 30
D  20 25 30  0

Starting city: A

Optimal Route Found:
A -> B -> D -> C -> A

Minimum Travel Cost: 80

Travelling Salesman Problem solved successfully!
```